

RAYAT BAHRA PROFESSIONAL UNIVERSITY



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NAME OF SUBJECT- WEB DESIGNING

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Unit B -Introduction to HTML and its Elements

Topic I- What is HTML?; HTML Documents; Basic structure of an HTML document;; Mark up Tags; Heading-Paragraphs; Line Breaks; HTML Tags.

What is HTML?;



- HTML stands for **Hyper Text Markup Language**
- HTML is the standard markup language for creating **Web pages**.
- HTML describes the **structure** of a **Web page**
- HTML elements tell the browser how to display the content
- HTML elements label pieces of content such as "this is a heading", "this is a paragraph", "this is a link", etc.
- As of 2025, over 95% of websites rely on HTML alongside CSS and JavaScript, making it a fundamental tool in modern web development.

What is HTML Continued...



- It is a markup language, not a programming language. This means it annotates text to define how it is structured and displayed by [web browsers](#).
- It is a **static language**, meaning it does not inherently provide interactive features but can be combined with [CSS](#) for styling and [JavaScript](#) for interactivity.

HTML Documents



An HTML document is a text-only file containing **Hypertext Markup Language** (HTML) code with a .html or .htm extension, used to structure and display content, such as text, images, and links, on a webpage. It uses tags enclosed in angle brackets (< >) to provide instructions to a web browser. HTML documents are also called web pages.

How it works:

You write HTML code in a text editor.
You save the file with the .html or .htm extension.
You open this file in a web browser.
The browser reads the HTML code, interprets the tags, and renders the structured content on the screen.

HTML Documents(A Simple HTML Document)



```
<!DOCTYPE html>
<html>
<head>
  <title>My First Webpage</title>
</head>
<body>
  <h1>Welcome to My Webpage</h1>
  <p>This is my first paragraph of text!</p>
</body>
</html>
```

Welcome to My Webpage

This is my first paragraph of text!

Basic structure of an HTML document

HTML Page Structure

`<!DOCTYPE html>` ← Tells version of HTML

`<html>` ← HTML Root Element

`<head>` ← Used to contain page HTML metadata

`<title>Page Title</title>` ← Title of HTML page

`</head>`

`<body>` ← Hold content of HTML

`<h2>Heading Content</h2>` ← HTML heading tag

`<p>Paragraph Content</p>` ← HTML paragraph tag

`</body>`

`</html>`

Basic structure of an HTML document



- <!DOCTYPE html> - This is the document type declaration, not a tag. It declares that the document is an HTML5 document.
- <html> - This is called the HTML root element. All other elements are contained within it.

Basic Structure continued..



- <head> - The head tag contains the "behind the scenes" elements for a webpage. Elements within the head aren't visible on the front end of a webpage.
- Typical elements inside the <head> include:
 - <title>: Defines the title displayed on the browser tab.
 - <meta>: Provides information like the character set or viewport settings.
 - <link>: Links external stylesheets or resources.
 - <style>: Embeds internal CSS styles.
 - <script>: Embeds JavaScript for functionality.
- <h2> - The <h2> tag is a second-level heading tag.
- <p> - The <p> tag represents a paragraph of text.
- <body> - The body tag is used to enclose all the visible content of a webpage. In other words, the body content is what the browser will show on the front end.

Creating an HTML document



- Steps to Create an HTML Document:
- **Open a Plain Text Editor:** Use a basic text editor such as Notepad (Windows), TextEdit (macOS), or a more advanced code editor like VS Code or Sublime Text. Avoid word processors like Microsoft Word, as they can add formatting that interferes with HTML.
- **Write the HTML Code:** Begin with the fundamental structure of an HTML document:

Creating an HTML document continued..



- **Save the File:**
- Go to "File" > "Save As..." in your text editor.
- Choose a location to save the file.
- Enter a file name, for example, index.html or myfirstpage.html.
- Crucially, ensure the file extension is .html or .htm. Some text editors might default to .txt, so you may need to manually change the "Save as type" or add the extension.

Creating an HTML document continued..



- **Open in a Web Browser:**
- Navigate to the location where you saved the HTML file.
- Double-click the file. It should open automatically in your default web browser (e.g., Chrome, Firefox, Edge).
- Alternatively, open your web browser and drag the HTML file directly into the browser window.

Mark up Tags;



- HTML (HyperText Markup Language) uses markup tags to structure and present content on the web. These tags are keywords enclosed in angle brackets, for example, `<html>`, `<body>`, `<p>`.
- **Paired Tags:**
- Most HTML tags come in pairs: an opening tag and a closing tag. The closing tag is identical to the opening tag but includes a forward slash before the tag name, for example, `<p>` (opening tag) and `</p>` (closing tag). Content is placed between these tags.
- **Self-Closing Tags/Singular Tags/Unpaired Tags:**
- Some tags are self-closing and do not require a separate closing tag because they don't enclose content. Examples include `<img>` for images and `<br>` for line breaks.



- **Attributes:**

- Tags can have attributes, which provide additional information or modify the behavior of the tag.
- Attributes are placed within the opening tag and typically consist of a name-value pair, for example,
- `<body bgcolor="red">`
- `<font size="20" color="yellow">`

- **Semantic Tags:**

- HTML5 introduced semantic tags (e.g., `<header>`, `<nav>`, `<article>`, `<footer>`) that provide meaning to the content they enclose, improving accessibility and search engine optimization.



- **Structure and Formatting:**
- Tags are used to define various elements of a web page, such as headings (<h1> to <h6>), paragraphs (<p>), lists (, ,), links (<a>), images (), and more, controlling how content is displayed and organized.



- **Types of HTML tags**
- **1. Based on Structure**
- **Paired Tags (Container Tags):**
- These tags require both an opening tag and a closing tag to define a content block. The closing tag includes a forward slash (/) before the tag name.
 - Example: `<p>This is a paragraph.</p>`
 - Other examples: `<div>`, `<span>`, `<h1>`, `<a>`, `<body>`
- **Unpaired Tags (Empty Tags/Void Elements):**
- These tags do not require a closing tag as they do not enclose content. They are used for self-contained elements.
 - Example: `<br>` (line break)
 - Other examples: `<img>`, `<hr>`



- **2. Based on Function/Purpose:**
- **Structural Tags:** Define the overall structure and sections of an HTML document.
 - Examples: `<html>`, `<head>`, `<body>`, `<div>`, `<span>`, `<header>`, `<footer>`, `<nav>`, `<article>`, `<section>`, `<aside>`
- **Text Formatting Tags:** Control the appearance and semantics of text.
 - Examples: `<h1>` to `<h6>` (headings), `<p>` (paragraph), `<b>` (bold), `<i>` (italic), `<strong>` (important text), `<em>` (emphasized text), `<sub>` (subscript), `<sup>` (superscript)
- **Link Tags:** Create hyperlinks to other resources.
 - Example: `<a>` (anchor tag)
- **Media Tags:** Embed images, audio, and video content.
 - Examples: `<img>`, `<audio>`, `<video>` etc.

Heading-Paragraphs



- HTML provides specific elements for defining headings and paragraphs to structure web content.
- Headings:
- HTML offers six levels of headings, from `<h1>` to `<h6>`.
- `<h1>` represents the most important heading (main title), while `<h6>` signifies the least important (sub-sub-heading).
- Headings are crucial for establishing a document's hierarchical structure, improving readability for users, and aiding search engine optimization (SEO) by indicating important keywords.

- `<!DOCTYPE html>`
- `<html>`
- `<body>`

- `<h1>Heading 1</h1>`
- `<h2>Heading 2</h2>`
- `<h3>Heading 3</h3>`
- `<h4>Heading 4</h4>`
- `<h5>Heading 5</h5>`
- `<h6>Heading 6</h6>`

- `</body>`
- `</html>`

Heading 1

Heading 2

Heading 3

Heading 4

Heading 5

Heading 6

Headings Are Important



- Search engines use the headings to index the structure and content of your web pages.
- Users often skim a page by its headings. It is important to use headings to show the document structure.
- `<h1>` headings should be used for main headings, followed by `<h2>` headings, then the less important `<h3>`, and so on.

HTML Paragraphs



- The HTML `<p>` element defines a paragraph.
- A paragraph always starts on a new line, and browsers automatically add some white space (a margin) before and after a paragraph.

```
<!DOCTYPE html>
<html>
<body>

<p>This is a paragraph.</p>
<p>This is a paragraph.</p>
<p>This is a paragraph.</p>

</body>
</html>|
```

This is a paragraph.

This is a paragraph.

This is a paragraph.



- Paragraphs:
- Paragraphs are defined using the `<p>` element.
- They are used to group related blocks of text, making content more organized and readable.
- Browsers typically render paragraphs with blank lines or indentation separating them from adjacent elements.
- `<p>`This is the first paragraph of text on the page. It contains several sentences explaining a particular topic.`</p>` `<p>`Here is a second paragraph, providing additional information or a different perspective on the subject.`</p>`

Relationship and Importance:



- Headings and paragraphs work together to create a clear and logical flow of information on a webpage.
- Headings provide an outline or summary of the content, while paragraphs fill in the details.
- Proper use of these elements enhances accessibility for users with screen readers and improves the overall user experience.

Line Breaks



- In HTML, line breaks are primarily inserted using the `<br>` tag. This tag creates a line break within a block of text, effectively moving the subsequent content to a new line without creating a new paragraph.
- `<p>`
- This is the first line.`<br>`
- This is the second line.`<br>`
- And this is the third line.
- `</p>`

Key points about the `<br>` tag:



- **Self-closing:**

- It is a void element, meaning it does not have a closing tag like `</br>`. In HTML5, `<br>` is the standard syntax. In XHTML, it might be written as `<br />`.

- **Purpose:**

- It is used for visual formatting, such as in addresses, poems, or lists of features where a new line is desired without the extra spacing or semantic meaning of a new paragraph.



- **Preformatted text:**
- For text where whitespace and line breaks should be preserved exactly as written in the source code, the `<pre>` tag can be used.
- `<pre>`
- This text
- will retain
- its formatting,
- including line breaks.
- `</pre>`

HTML Tags



```
<HTML>
<HEAD>
<TITLE>PRACTICAL NO.1</TITLE>
</HEAD>
<BODY >
<pre>
<B>BOLD</B>
<U>UNDERLINE</U>
<I>ITALICS</I>
<STRONG>STRONG TEXT</STRONG>
<DEL>DELETE</DEL>
<INS>INSERT</INS>
<H1>HEADING 1</H1>
<H2>HEADING 2</H2>
<H3>HEADING 3</H3>
<H4>HEADING 4</H4>
<H5>HEADING 5</H5>
<H6>HEADING 6</H6>
>
```



h₂o

<hr

a²b

<hr>

<center>CENTER</center>

<p>paragraph tag</p>

FONT tag

<mark>highlighted tag</mark>

</pre></body>

</html>

BOLD
UNDERLINE
ITALICS
STRONG TEXT
~~DELETE~~
INSERT

HEADING 1

HEADING 2

HEADING 3

HEADING 4

HEADING 5

HEADING 6

h₂o

a²b

CENTER

paragraph tag

FONT tag

Highlighted tag